

I/O-efficient Symbolic Model Checking

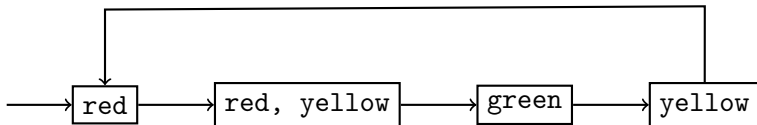
Steffan Christ Sølvesten

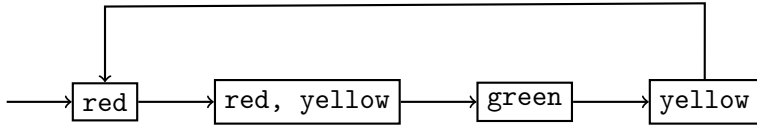
15th of May 2025



defer (difœ').

defer (difœ'). *v.t.* to put off; to postpone.
[...]

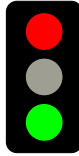






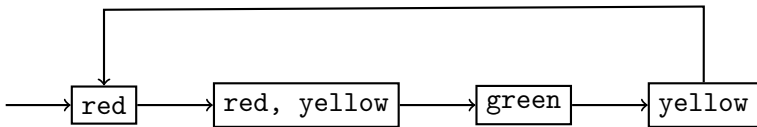






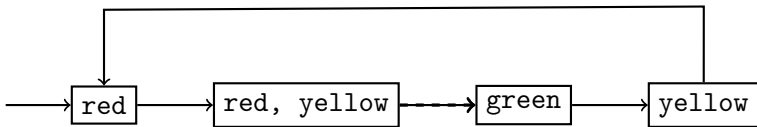
"It is either red and/or ($\vee$) yellow or it is exclusively ($\oplus$) green."

$(\text{red} \vee \text{yellow}) \oplus \text{green}$

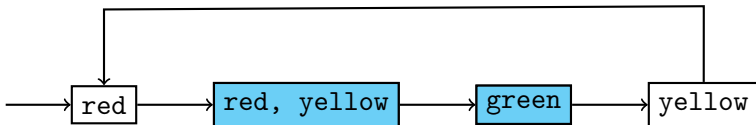


"When red and yellow, it turns exclusively green."

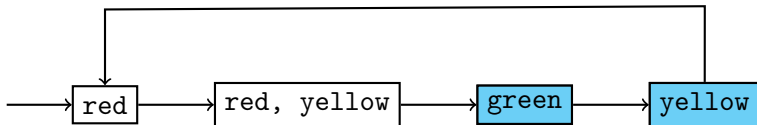
$$\text{red} \wedge \text{yellow} \rightarrow \neg \text{red}' \wedge \neg \text{yellow}' \wedge \text{green}'$$



$$RelProd(S_{\vec{x}}, T_{\vec{x}, \vec{x}'}) \triangleq (\exists \vec{x} . S_{\vec{x}} \wedge T_{\vec{x}, \vec{x}'}) [\vec{x}' \mapsto \vec{x}]$$

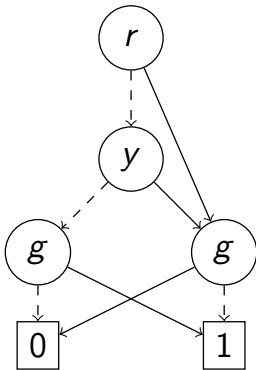


$$RelProd(S_{\vec{x}}, T_{\vec{x}, \vec{x}'}) \triangleq (\exists \vec{x} . S_{\vec{x}} \wedge T_{\vec{x}, \vec{x}'}) [\vec{x}' \mapsto \vec{x}]$$



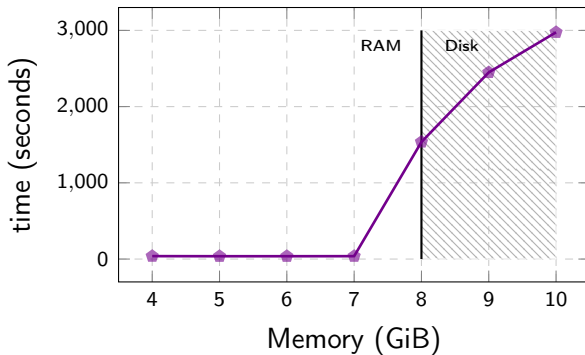
Binary Decision Diagram (BDD)

$$(\text{red} \vee \text{yellow}) \oplus \text{green}$$



Usually BDDs are implemented
by means of:

- Depth-first recursion
- Hash Tables



BuDDy BDD Library

- **Formal Methods:**

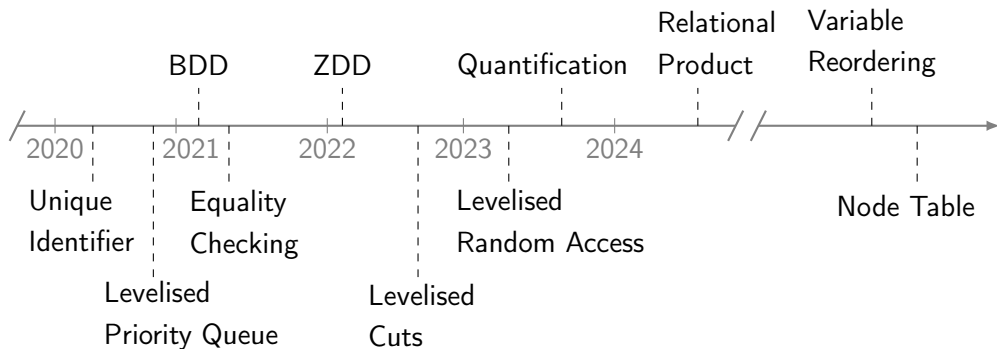
Motivation and applications of algorithms.

- **Algorithmics:**

Theoretical tools for the design and analysis of algorithms.

- **Algorithm Engineering:**

Experimental evaluation and design for real-life computers.



The I/O-Complexity of Ordered Binary-Decision Diagram Manipulation

Lars Arge

Department of Computer Science

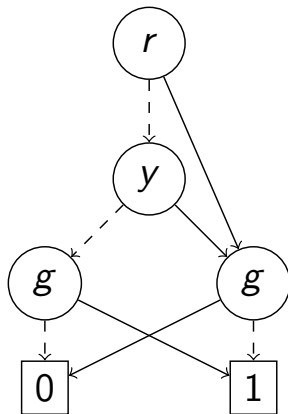
University of Aarhus

August 1996

Processing Order

■ Depth-first:

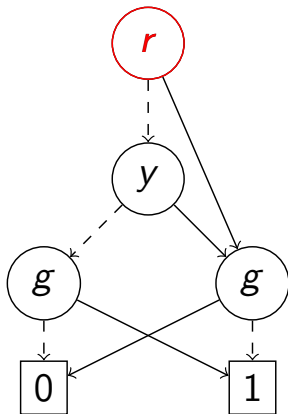
Last in, first out Queue (Stack)



Processing Order

■ Depth-first:

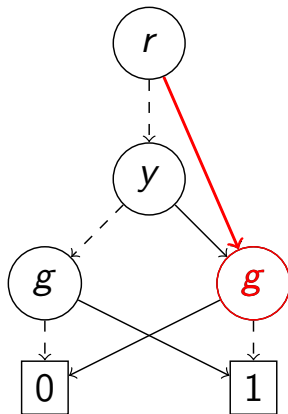
Last in, first out Queue (Stack)



Processing Order

■ Depth-first:

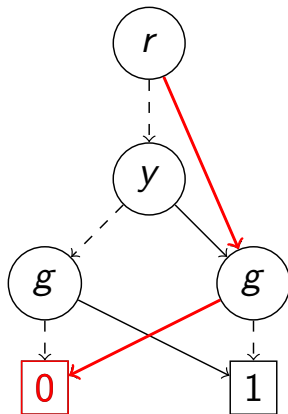
Last in, first out Queue (Stack)



Processing Order

■ Depth-first:

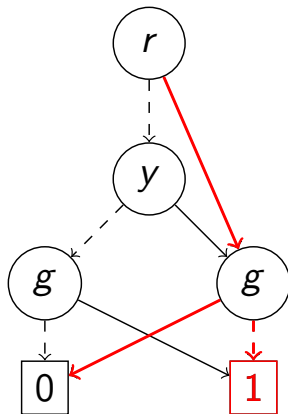
Last in, first out Queue (Stack)



Processing Order

■ Depth-first:

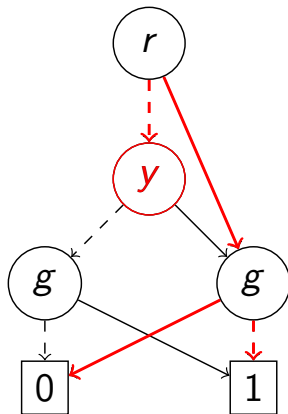
Last in, first out Queue (Stack)



Processing Order

■ Depth-first:

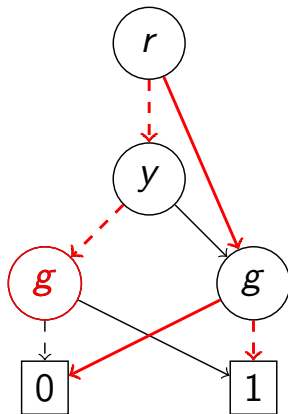
Last in, first out Queue (Stack)



Processing Order

■ Depth-first:

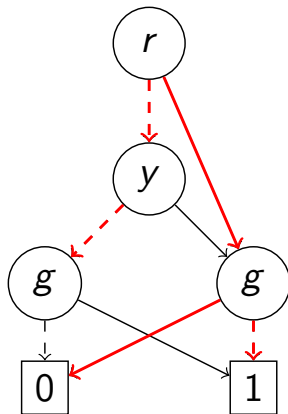
Last in, first out Queue (Stack)



Processing Order

■ Depth-first:

Last in, first out Queue (Stack)



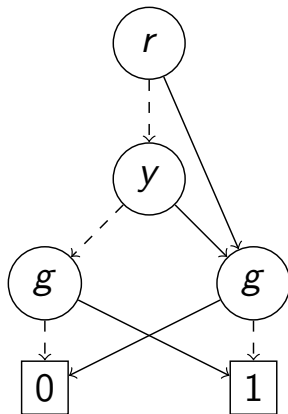
Processing Order

- **Depth-first:**

Last in, first out Queue (Stack)

- **Breadth-first:**

First in, first out Queue



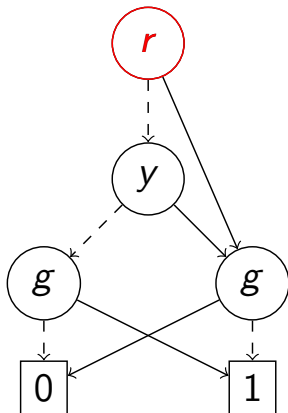
Processing Order

- **Depth-first:**

Last in, first out Queue (Stack)

- **Breadth-first:**

First in, first out Queue



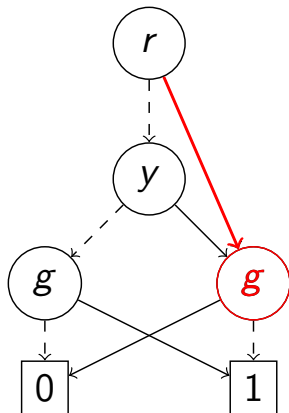
Processing Order

■ Depth-first:

Last in, first out Queue (Stack)

■ Breadth-first:

First in, first out Queue



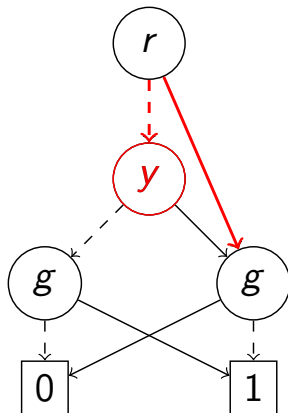
Processing Order

■ Depth-first:

Last in, first out Queue (Stack)

■ Breadth-first:

First in, first out Queue



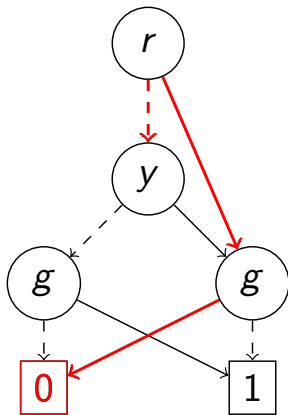
Processing Order

■ Depth-first:

Last in, first out Queue (Stack)

■ Breadth-first:

First in, first out Queue



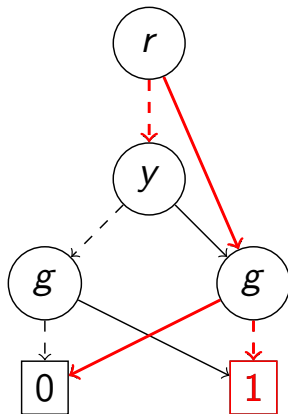
Processing Order

■ Depth-first:

Last in, first out Queue (Stack)

■ Breadth-first:

First in, first out Queue



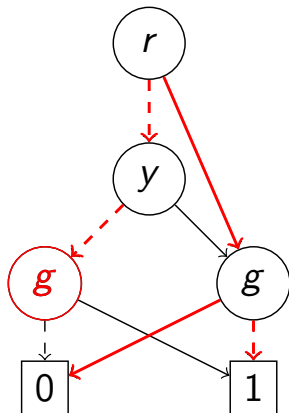
Processing Order

■ Depth-first:

Last in, first out Queue (Stack)

■ Breadth-first:

First in, first out Queue



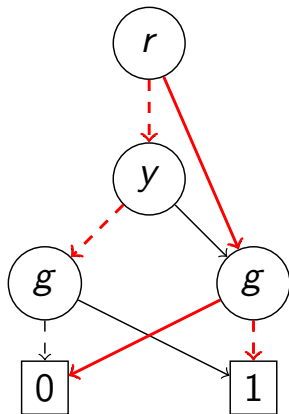
Processing Order

■ Depth-first:

Last in, first out Queue (Stack)

■ Breadth-first:

First in, first out Queue



Processing Order

- **Depth-first:**

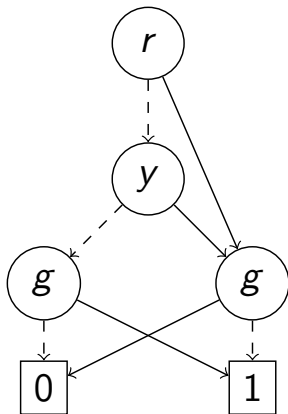
Last in, first out Queue (Stack)

- **Breadth-first:**

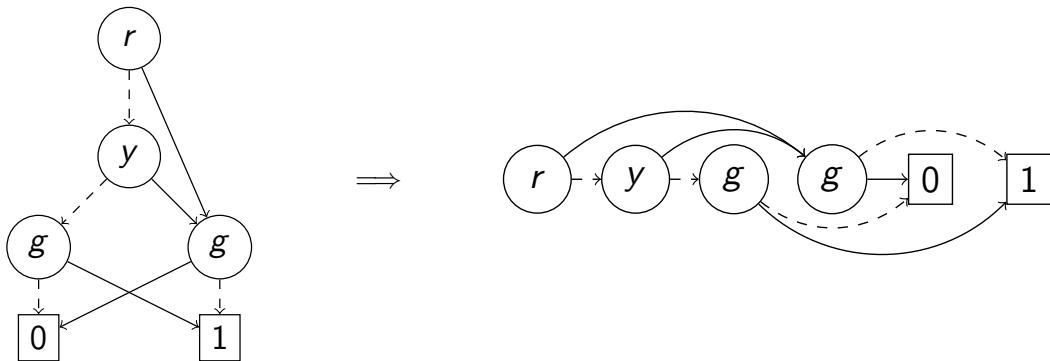
First in, first out Queue

- **Time-forward Processing:**

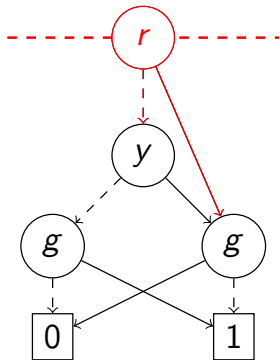
Priority Queue



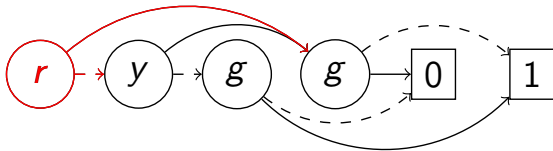
Time-forward Processing



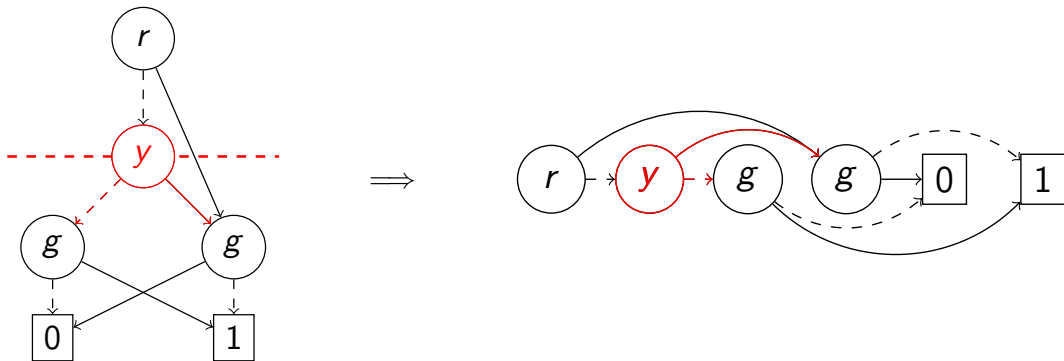
Time-forward Processing



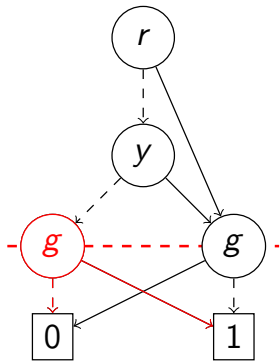
$\Rightarrow$



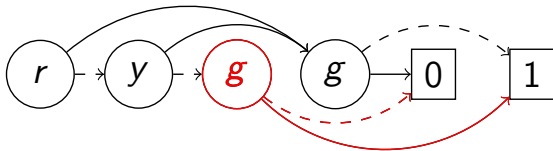
Time-forward Processing



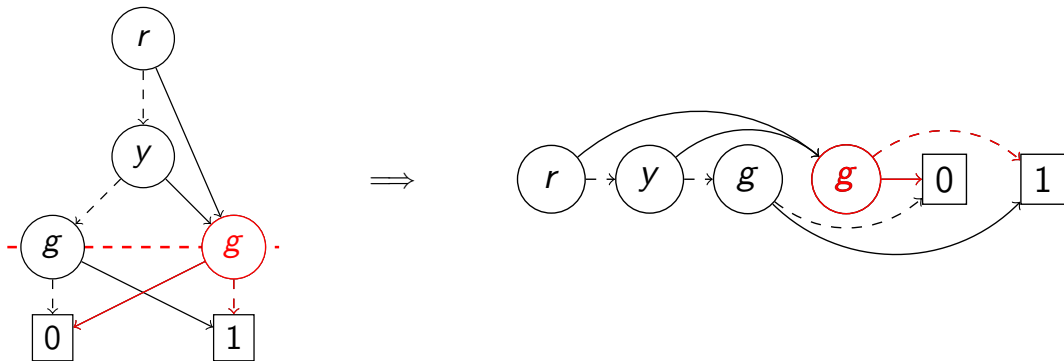
Time-forward Processing



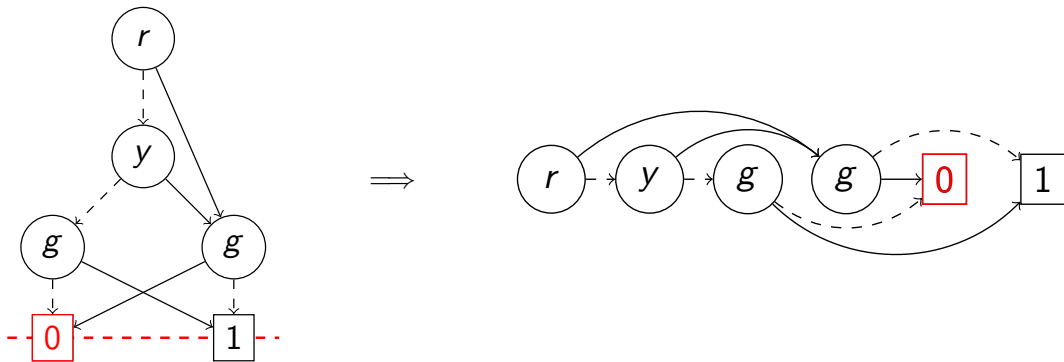
$\Rightarrow$



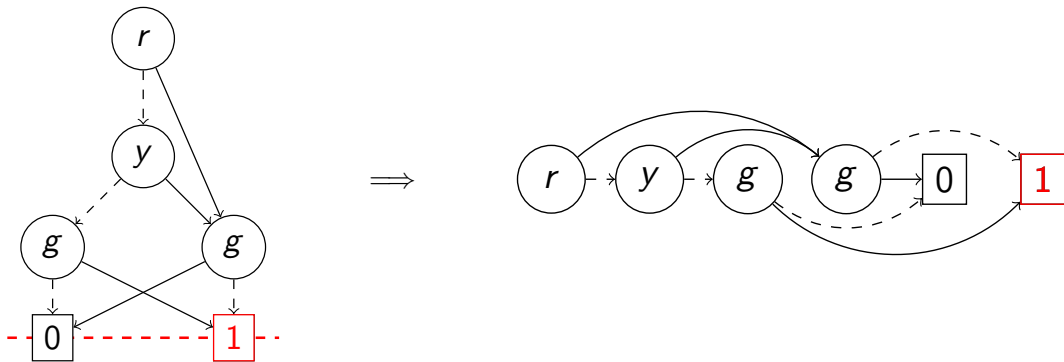
Time-forward Processing



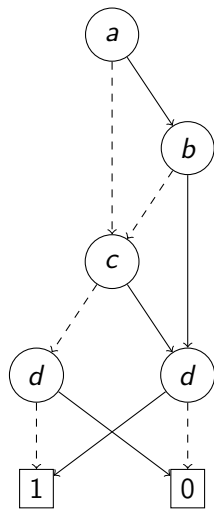
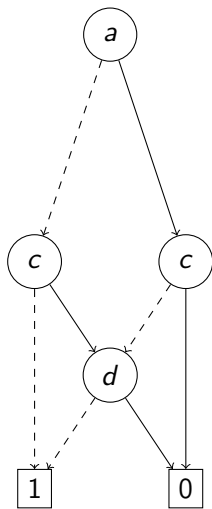
Time-forward Processing



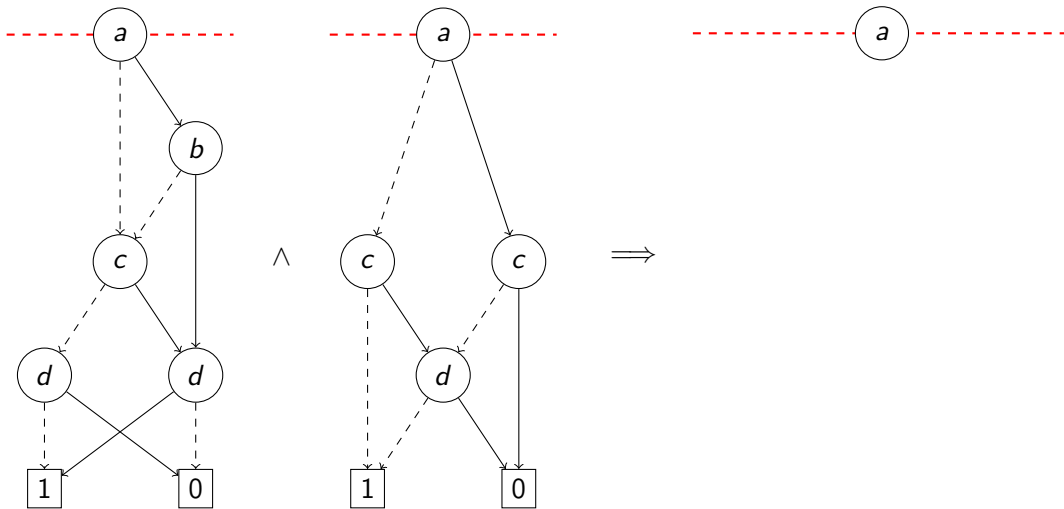
Time-forward Processing

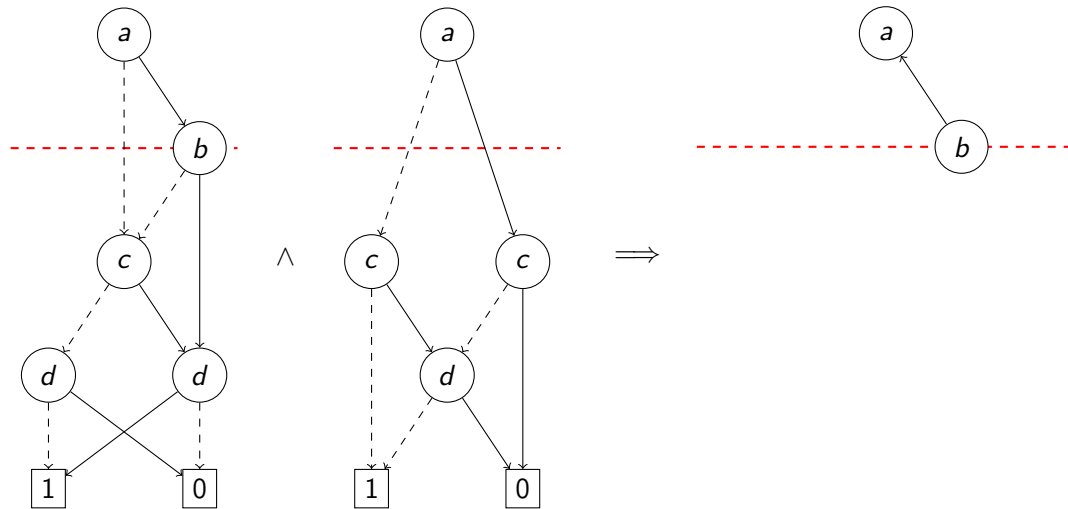


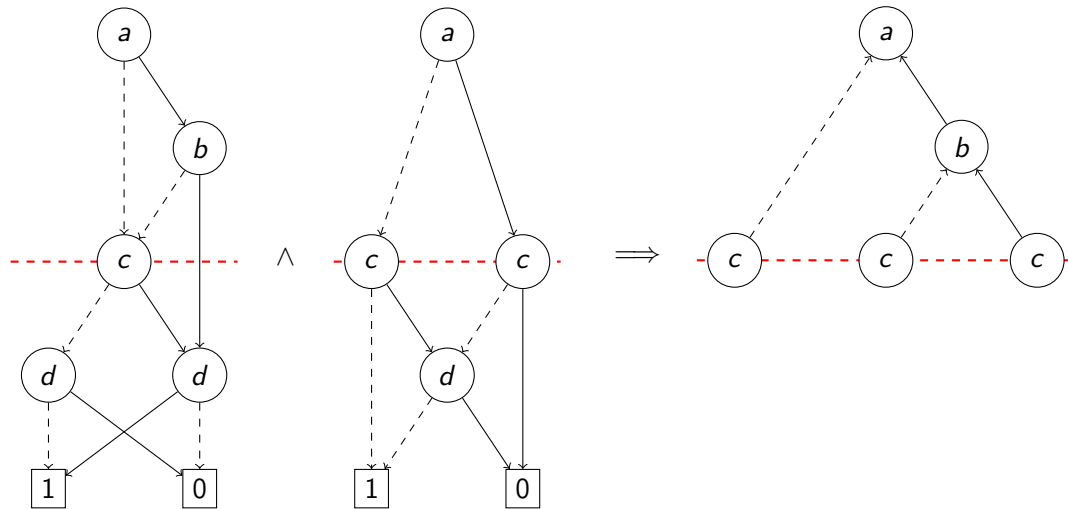
$$\phi \wedge \psi$$

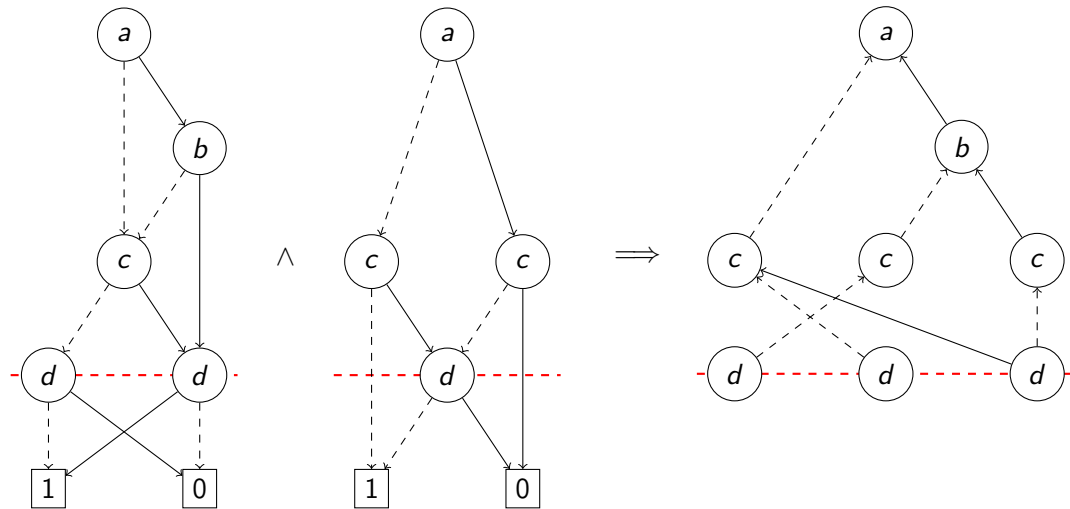
$\phi \wedge \psi$

 $\wedge$

 $\Rightarrow$

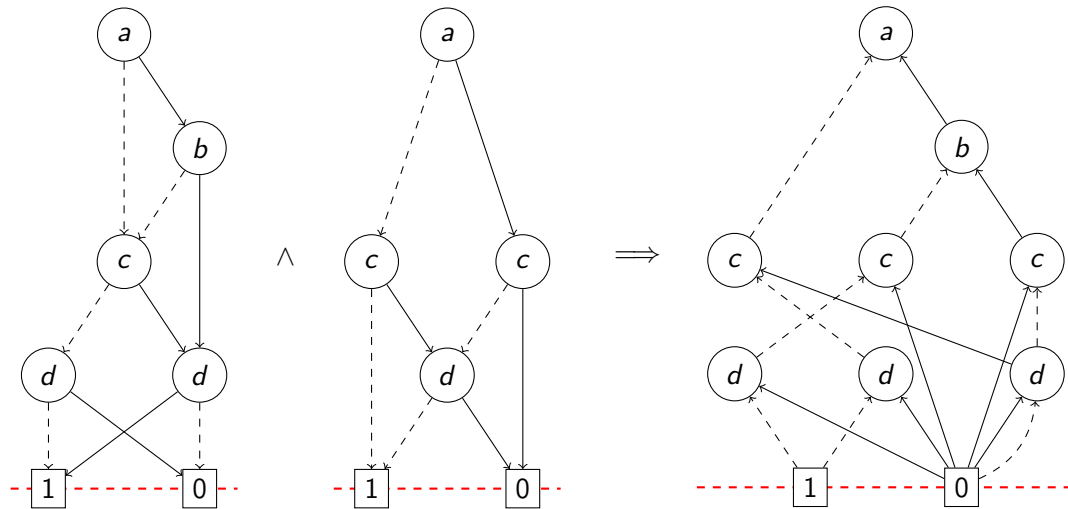
$\phi \wedge \psi$

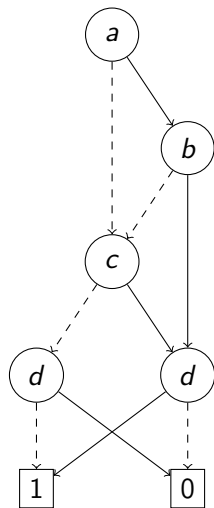
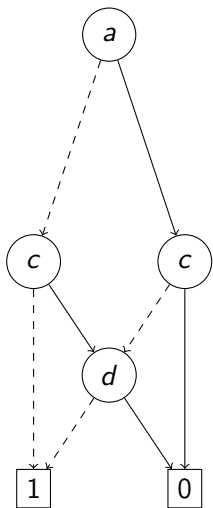
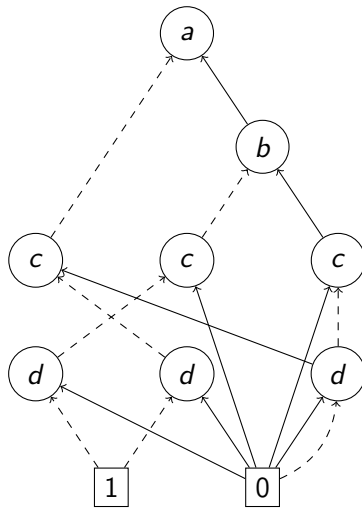


$\phi \wedge \psi$


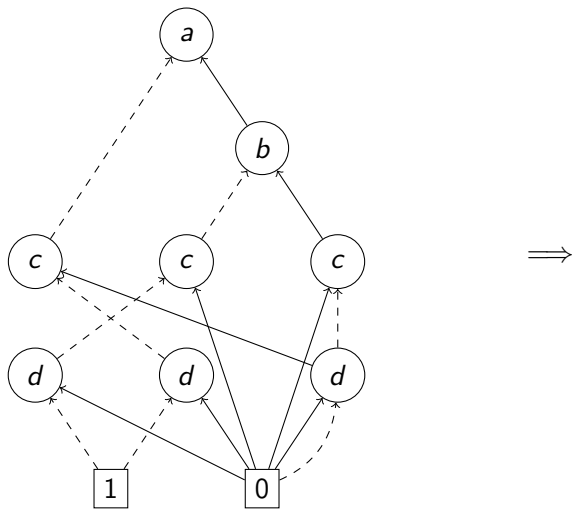
$\phi \wedge \psi$


$\phi \wedge \psi$


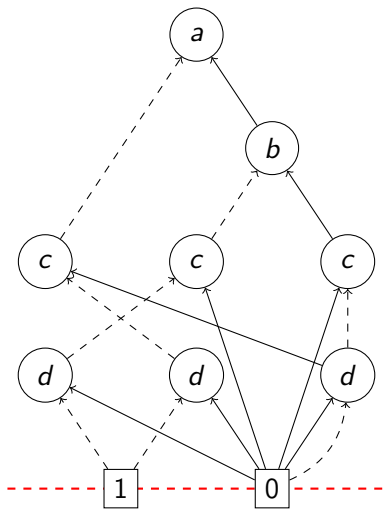
$\phi \wedge \psi$


$\phi \wedge \psi$

 $\wedge$

 $\Rightarrow$


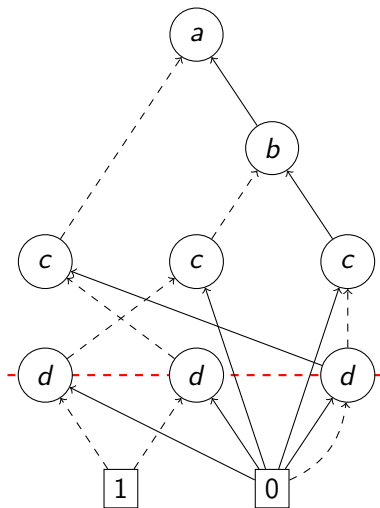
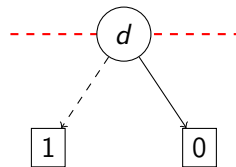
$$\phi \wedge \psi$$



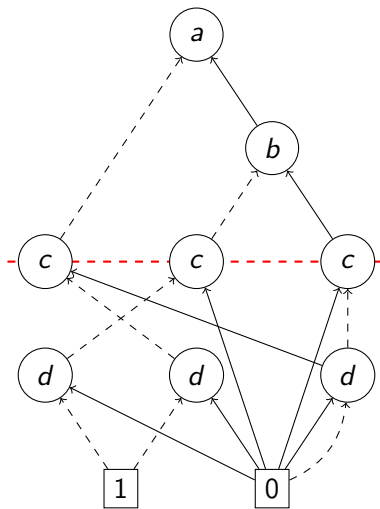
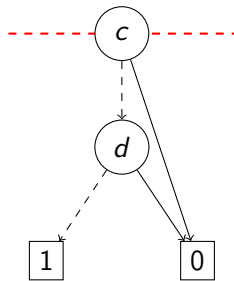
$$\phi \wedge \psi$$


 $\Rightarrow$

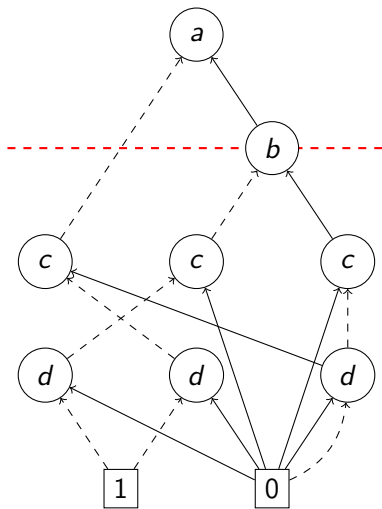
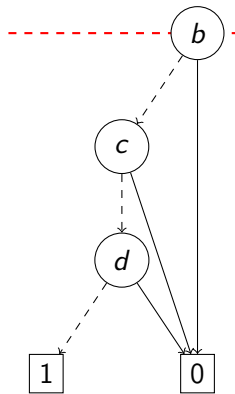

$$\phi \wedge \psi$$

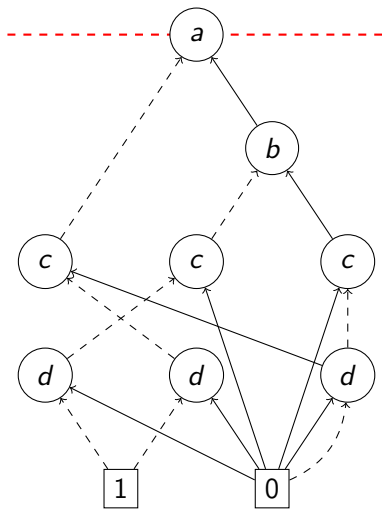
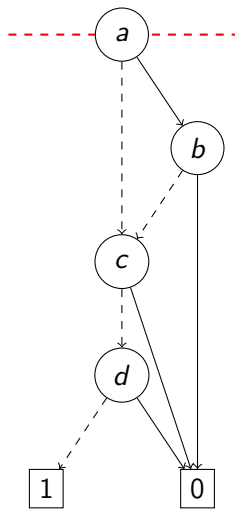

 $\Rightarrow$


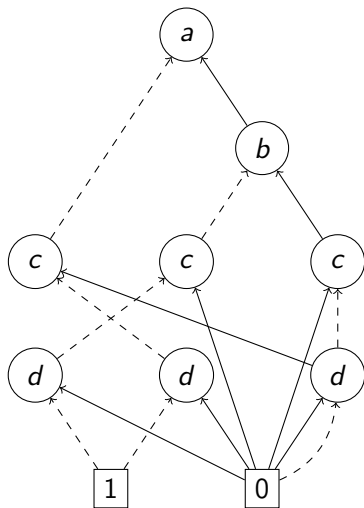
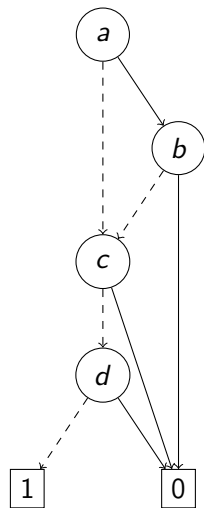
$$\phi \wedge \psi$$


 $\Rightarrow$


$$\phi \wedge \psi$$


 $\Rightarrow$


$\phi \wedge \psi$

 $\Rightarrow$


$\phi \wedge \psi$

 $\Rightarrow$


$$\phi \wedge \psi$$

Depth-first

$$\mathcal{O}(N + T)$$

but not I/O efficient!

Time-forward Processing

$$\mathcal{O}((N + T) \log(N + T))$$

but I/O efficient!

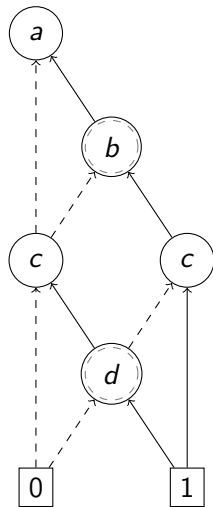
N : Input Size

T : (Unreduced) Output Size

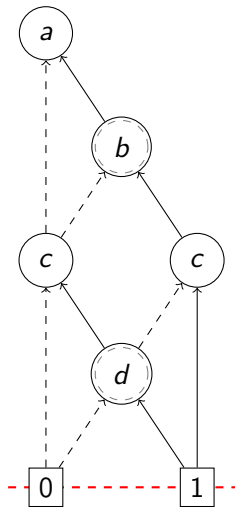
$$\exists x : \phi(x)$$

$$\exists x : \phi(x) \equiv \phi(0) \vee \phi(1)$$

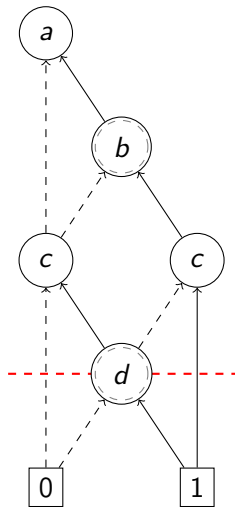
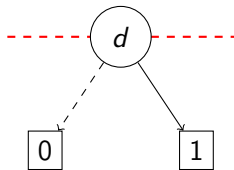
$$\exists \vec{x} : \phi(\vec{x}) \quad \vec{x} = \{b, d\}$$


 $\Rightarrow$

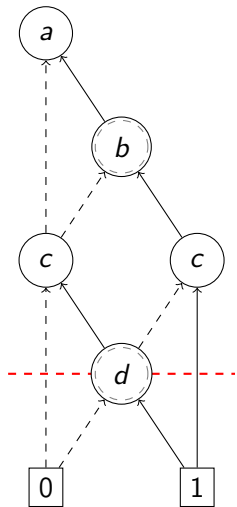
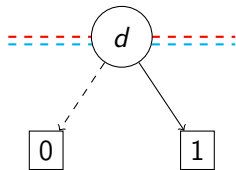
$$\exists \vec{x} : \phi(\vec{x}) \quad \vec{x} = \{b, d\}$$


 $\Rightarrow$

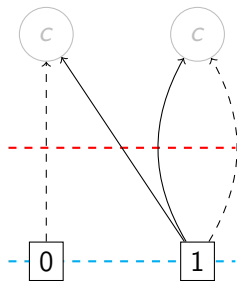
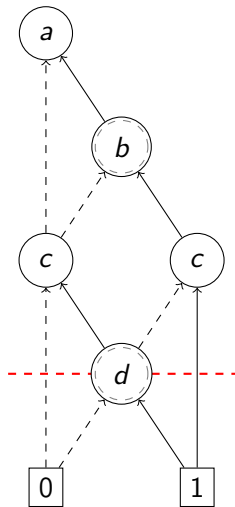
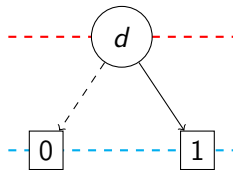

$$\exists \vec{x} : \phi(\vec{x}) \quad \vec{x} = \{b, d\}$$


 $\Rightarrow$


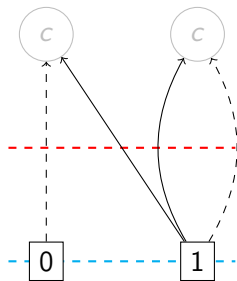
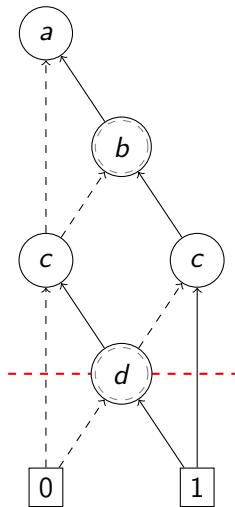
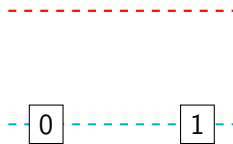
$$\exists \vec{x} : \phi(\vec{x}) \quad \vec{x} = \{b, d\}$$


 $\Rightarrow$


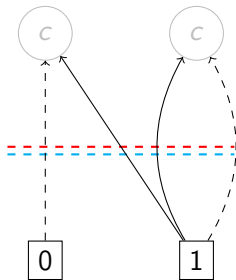
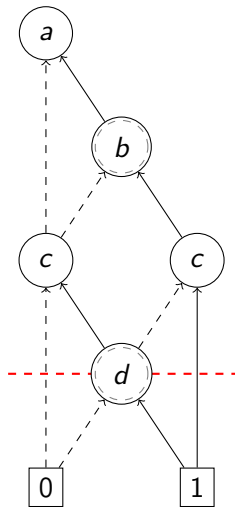
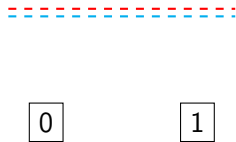
$$\exists \vec{x} : \phi(\vec{x}) \quad \vec{x} = \{b, d\}$$


 $\Rightarrow$


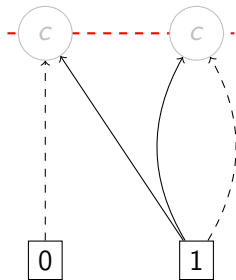
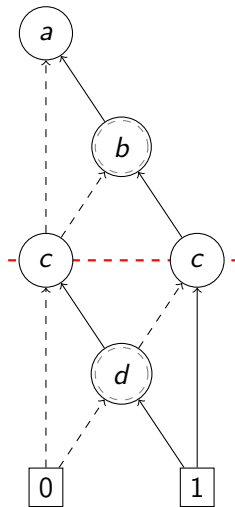
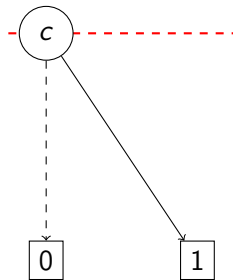
$$\exists \vec{x} : \phi(\vec{x}) \quad \vec{x} = \{b, d\}$$


 $\Rightarrow$


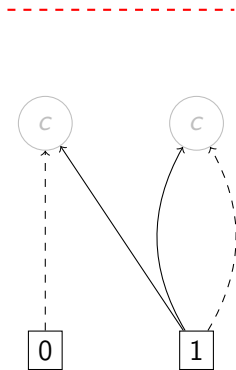
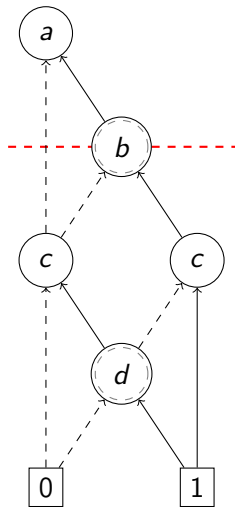
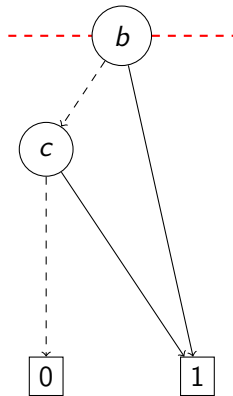
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 $\Rightarrow$


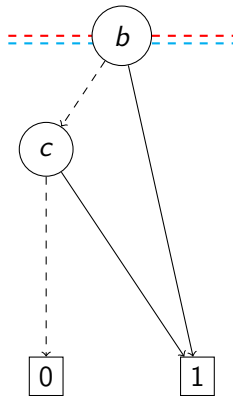
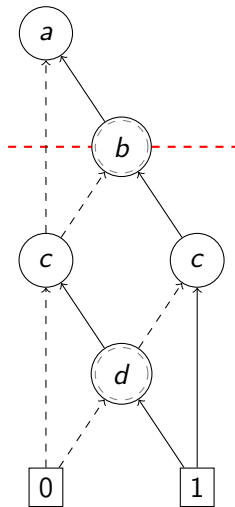
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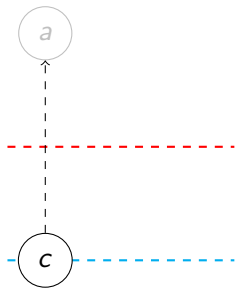
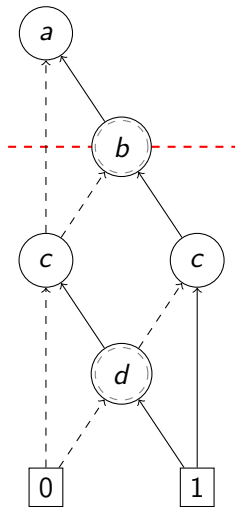
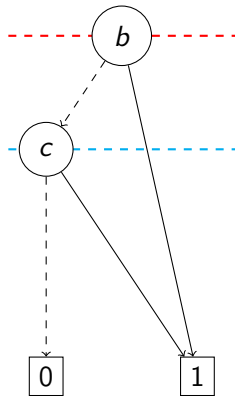
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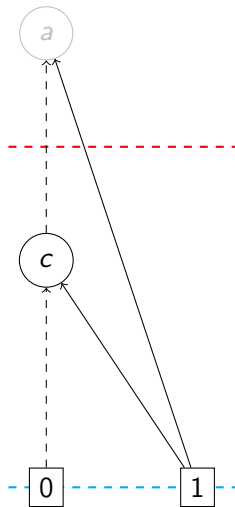
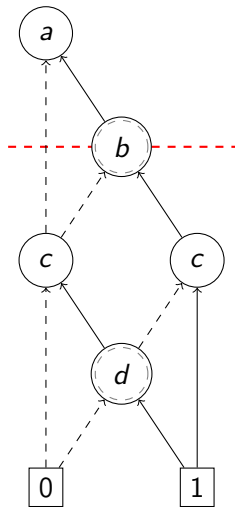
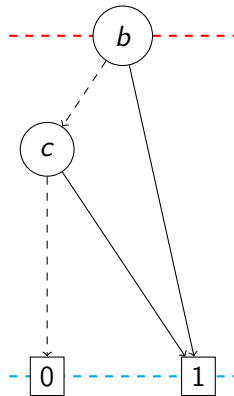
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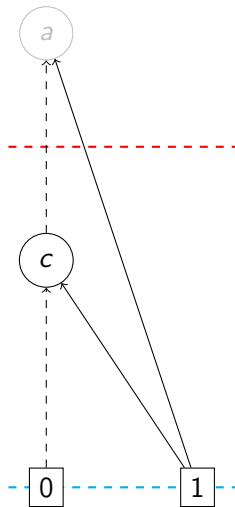
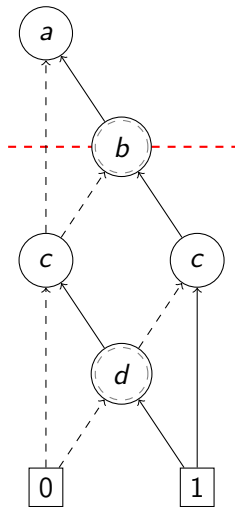
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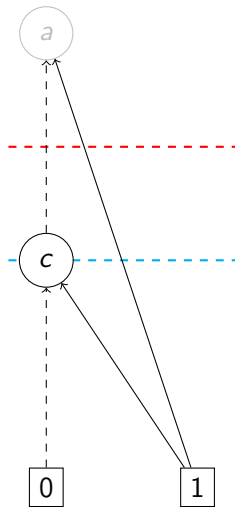
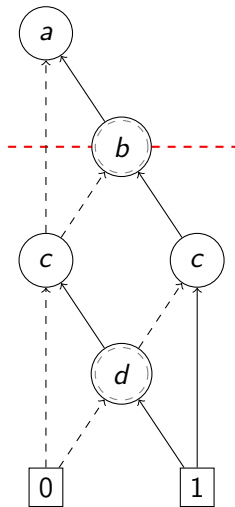
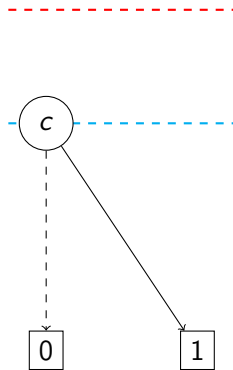
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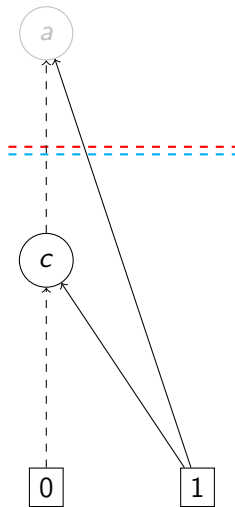
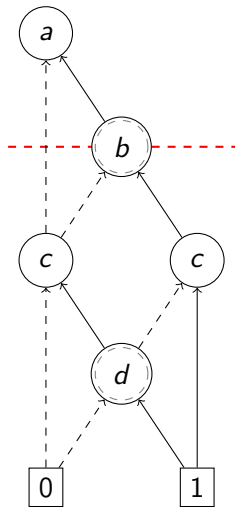
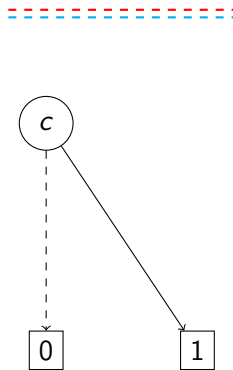
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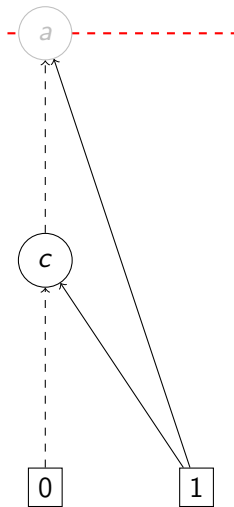
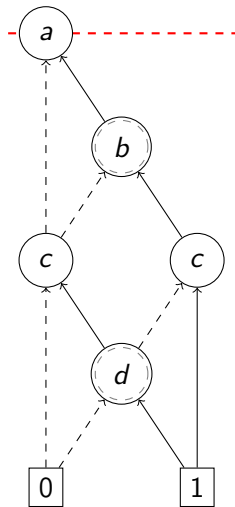

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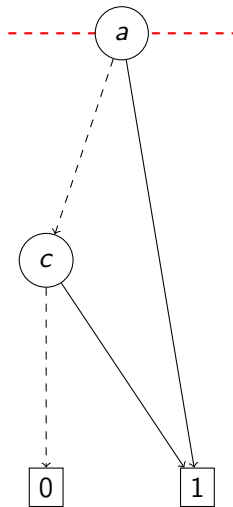
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 $\Rightarrow$


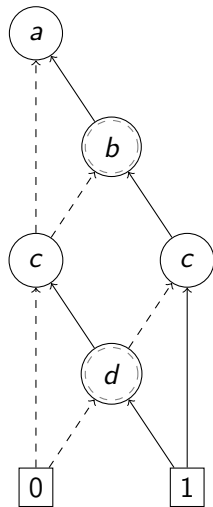
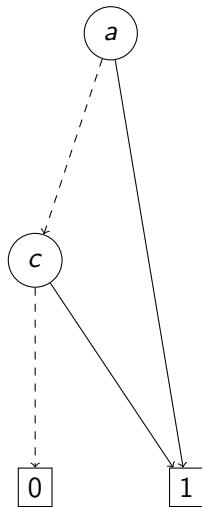
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$\Rightarrow$



$$\exists \vec{x} : \phi(\vec{x}) \quad \vec{x} = \{b, d\}$$


 $\Rightarrow$


$$\exists \vec{x} : \phi(\vec{x})$$

Single Variable Quantification

$$\mathcal{O}(N^{2^k} \log(N^{2^k}))$$

Nested Sweeping

$$\mathcal{O}(N^{2^k} \log(N^{2^k}))$$

but 1.8× faster!

N : Input Size

$$[x \mapsto y]$$

$$[x \mapsto y]$$

Definition

A relabelling π is monotone if $x_i < x_j \implies \pi(x_i) < \pi(x_j)$.

Lemma

If π is monotone, then the BDD $f(\vec{x})$ is isomorphic to $f(\pi(\vec{x}))$.



$$[x \mapsto y]$$

Definition

A relabelling π is monotone if $x_i < x_j \implies \pi(x_i) < \pi(x_j)$.

Lemma

If π is monotone, then the BDD $f(\vec{x})$ is isomorphic to $f(\pi(\vec{x}))$.

- **One can apply π in a single linear scan.**
 $\mathcal{O}(N)$ time, $2 \cdot \text{scan}(N)$ I/Os, and N external space.
- **One can incorporate π into a (succeeding) top-down sweep.**
 $\mathcal{O}(N)$ time, 0 I/Os, and 0 external space.
- **One can incorporate π into a (preceeding) bottom-up sweep.**
 $\mathcal{O}(n)$ time, 0 I/Os, and 0 external space.

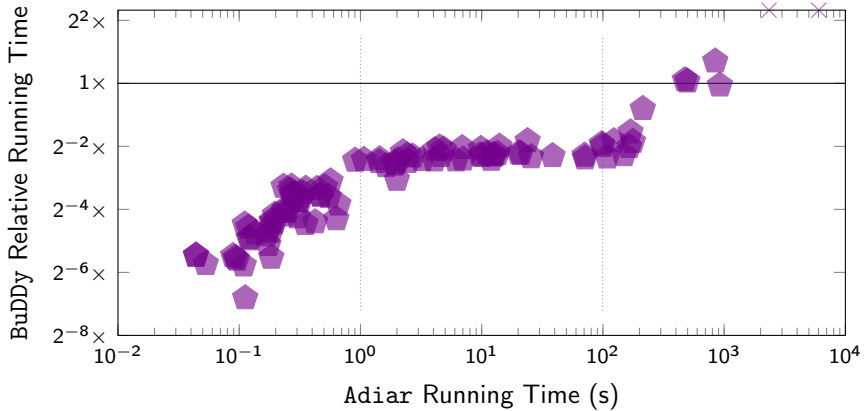
Adiar¹

github.com/ssoelvsten/adiar

¹ adiar ⟨ portugese ⟩ (verb) : to defer, to postpone.

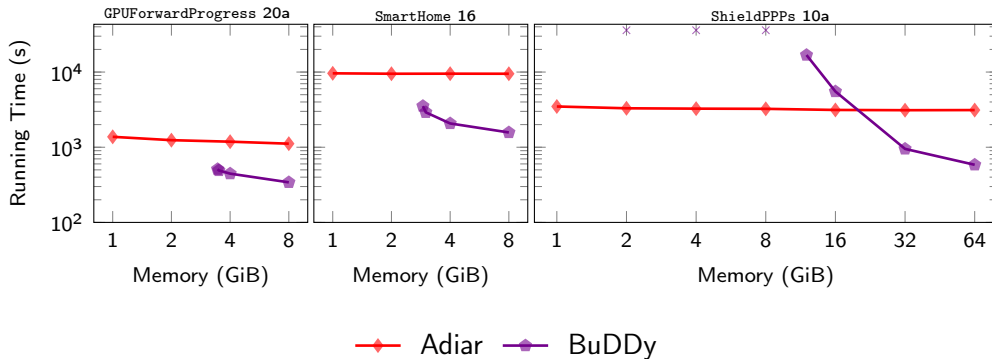
Quantified Boolean Formulæ (QBF) of 2-player games:

$$\exists \vec{x} \forall \vec{y} \dots \exists \vec{z} : \phi(\vec{x}, \vec{y}, \dots, \vec{z}) \stackrel{?}{=} 1$$



Relational Product in a Transition System:

$$RelProd(S_{\vec{x}}, T_{\vec{x}, \vec{x}'}) \triangleq (\exists \vec{x} . S_{\vec{x}} \wedge T_{\vec{x}, \vec{x}'}) [\vec{x}' \mapsto \vec{x}]$$



Adiar

📄 github.com/ssoelvsten/adiar

⚖ MIT

📄 ssoelvsten.github.io/adiar

✓ 3.462 unit tests

Adiar

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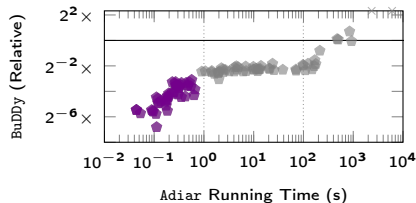
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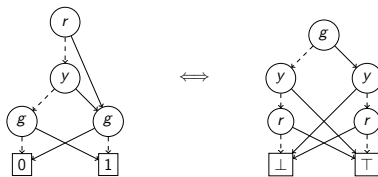
✓ 3.462 unit tests

Future Work

 Small Instances



 Variable Reordering



Manual Variable Reordering

Consider the $\text{bdd_replace}(f, \pi)$ from BuDDy to compute $[x \mapsto \pi(x)]$ on an input with N BDD nodes and n variables and an output of T BDD nodes.

		Depth-first	Time-forwarding
Any π	Time & I/Os Space	$\mathcal{O}(N + n \cdot T)$	$\mathcal{O}(\text{sort}(T \cdot \sum_{i=1}^n C_{1:f[i]}^{\emptyset}))$ $\mathcal{O}(\text{sort}(T \cdot \max_i(C_{1:f[i]}^{\emptyset})))$
Exchange	Time & I/Os Space	$\mathcal{O}(N + T)$	$\mathcal{O}(\text{sort}(N + T))$ $\mathcal{O}(N + T)$
Adjacent Swap	Time & I/Os Space	$\mathcal{O}(N + T)$	$\mathcal{O}(\text{sort}(N + T))$ $\mathcal{O}(N + T)$

Dynamic Variable Reordering

We have surveyed current dynamic variable ordering methods to uncover how our I/O-efficient manual reordering algorithms can be applied.

Metaheuristics: simulated annealing, genetic and memetic algorithms, and swarm intelligence algorithms, via *exchanges* and *adjacent swaps*¹.

Sifting: Rudell's sifting algorithm, via repeated *exchanges*.

Parallel Sifting: Rudell's procedure via repeated *adjacent swaps*; this is akin to the 2-window algorithm.

In terms of space, these I/O-efficient variants are on par with the depth-first approach.

¹Or any *non-monotone* π if that does not break the memory limits.

- **Unique Identifier:**

Sorting predicates can be turned into mere 64-bit integer comparison.

- **Levelised Priority Queue:**

Defer sorting of level ℓ in the priority queue until level ℓ has to be processed.

- **Equality Checking:**

If BDDs ϕ and ψ are created by Reduce then they are bit-wise equivalent iff $\phi \equiv \psi$.

- **Levelised Cuts:**

The priority queue's size is at most the maximum cut in the BDD.

- **Levelised Random Access:**

If a BDD's level fits into memory then random access can be used (in moderation).

- **Node Table:**

If a BDD is small enough then compute on it with the conventional approach.

Unique Identifier



(a) null.

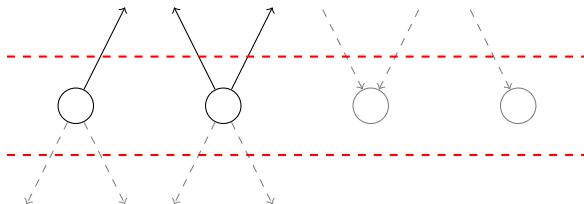


(b) leaf with value 0 or 1.



(c) node with label, i, identifier, id, and an out-degree index, o.

Levelised Priority Queue



Observation: When processing level i , no new requests for the same level are made.

Optimisation: Sort bucket of *all* requests for level i at once with Quicksort ($\sim 2\times$ faster than a priority queue).

	 Improvement (%)
Queens (14)	25.3
Tic-Tac-Toe (22)	37.0

Equality Checking

$\mathcal{O}(\text{sort}(N^2))$: compute $f \leftrightarrow g$ and check whether it is the 1 BDD.

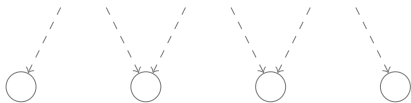
$\mathcal{O}(\text{sort}(N))$: Fail-fast during a product construction if more than $N_{f,i}$ ($N_{g,i}$) pairs of nodes are checked on level i .

$2 \cdot \text{scan}(N)$: Fail-fast during a linear scan of both BDDs bit-by-bit.

	
	Time (s)
$\mathcal{O}(\text{sort}(N^2))$	0.38
$\mathcal{O}(\text{sort}(N))$	0.058
$2 \cdot \text{scan}(N)$	0.006

Checking the (EPFL Benchmark)
voter circuit's single output gate
($|N_f| = |N_g| = 5.76$ MiB).

Levelised Cuts



Theorem

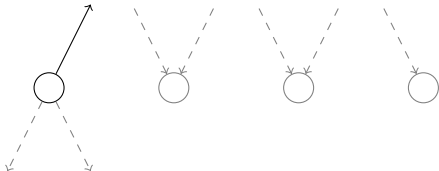
Given maximum 2-level cuts size C_ϕ for BDD ϕ and C_ψ for BDD ψ , the maximum 2-level cut for the BDD $\phi \wedge \psi$ is less than or equal to $C_\phi \cdot C_\psi$.

Lemma

The maximum 2-level cut for BDD ϕ is at most $\frac{3}{2}$ larger than its maximum 1-level cut.

		+🕒	📏
		Overhead	Precision
1-level cut	:	1.0%	69.2%
2-level cut	:	3.3%	86.3%

Levelised Cuts



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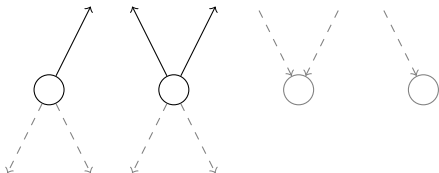
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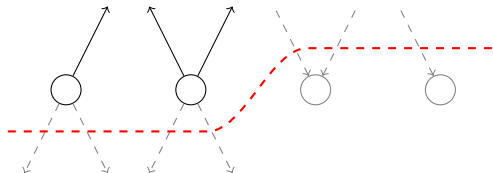
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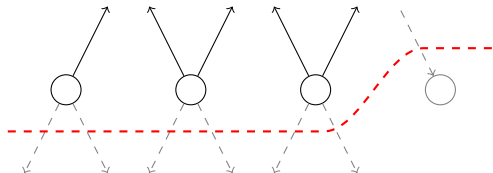
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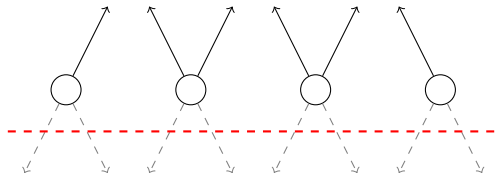
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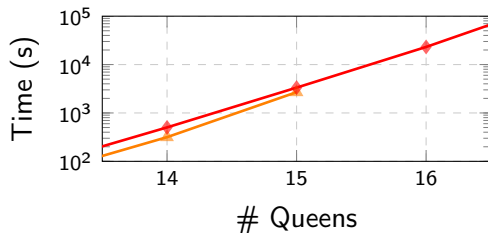
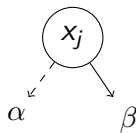
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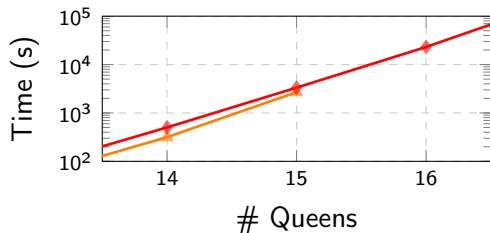
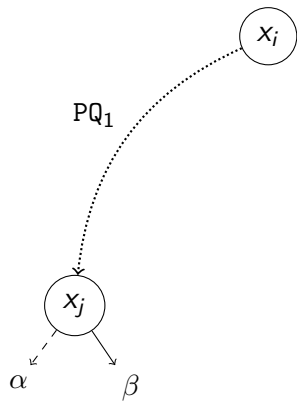
Levelised Random Access



🕒				
△	CUDD	v3.0	:	44.8 min
◇	Adiar	v1.0	:	66.7 min
	+ cuts		:	56.8 min

Counting solutions for the N-Queens Problem.

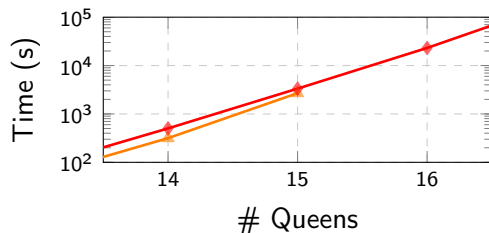
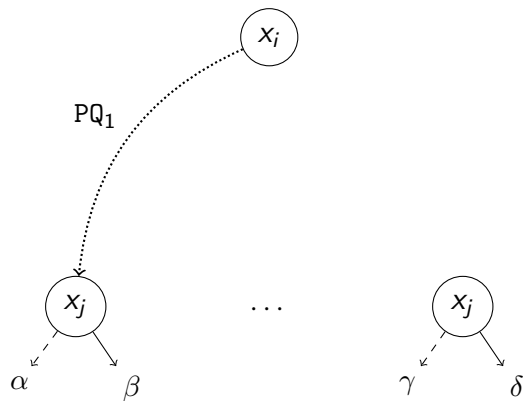
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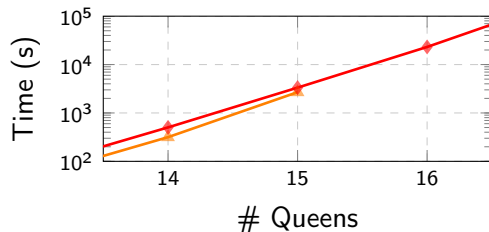
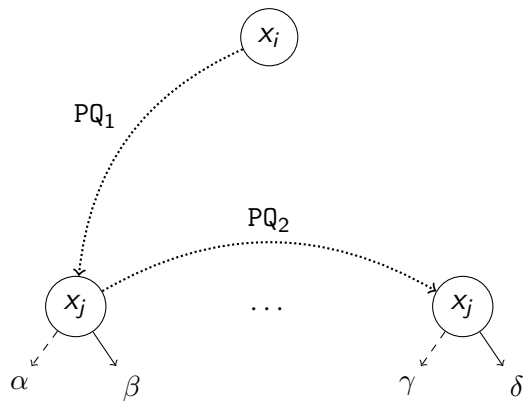
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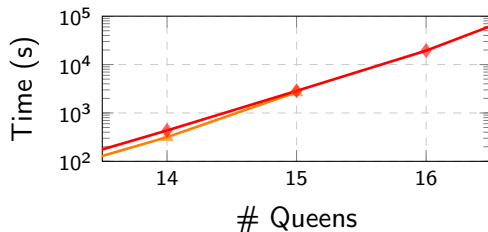
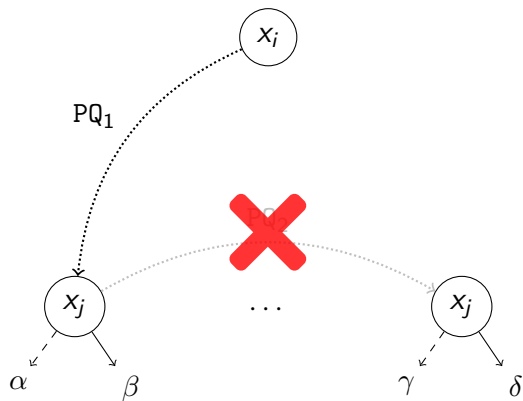
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Levelised Random Access



				🕒
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◇	Adiar	v1.0	:	66.7 min
	+ cuts		:	56.8 min
	+ random access		:	47.2 min

Counting solutions for the N-Queens Problem.

Node Table

